1984ApJ...280..465L

THE ASTROPHYSICAL JOURNAL, 280:465–469, 1984 May 15
© 1984. The American Astronomical Society. All rights reserved. Printed in U.S.A.

MOLECULES IN THE EARLY UNIVERSE

STEPHEN LEPP[1] AND J. MICHAEL SHULL[2]
Joint Institute for Laboratory Astrophysics, University of Colorado and National Bureau of Standards
Received 1983 September 9; accepted 1983 November 23

ABSTRACT

We present calculations of the formation of astrophysically interesting molecules (H_2, HD, LiH, and HeH^+) by gas-phase reactions during the postrecombination epoch (redshifts $z = 300$–30). In standard Friedmann cosmological models, $H_2/H \approx 10^{-6}$, $HD/H_2 \approx 10^{-4.5}$, and $LiH/H_2 \approx 10^{-6.5}$. These molecules may dominate the cooling and trigger the collapse of primordial gas clouds. The dipole rotational transitions of HD and LiH are particularly important at high density and low temperature. Additional molecules form during spherical collapse of these clouds, and their rotational cooling keeps the gas temperature between 400 and 1500 K over 12 decades of density increase until the H_2 lines become optically thick. The existence of molecular coolants at high redshift has significant implications for the first generation of stars and for thermal instabilities in intergalactic matter.

Subject headings: cosmology — early universe — interstellar: molecules

I. INTRODUCTION

If one accepts the conventional model of nucleosynthesis in the "hot big bang" (Peebles 1966; Wagoner, Fowler, and Hoyle 1967), the first stars are formed from gas consisting of hydrogen (^1H), helium (^4He), and traces of deuterium (^2H), ^3He, and ^7Li. In the absence of heavy elements or dust grains, the processes of radiative cooling, cloud collapse, and star formation were undoubtedly quite different from those today. For example, cooling by Lyα is ineffective for $T < 8000$ K, well above the matter temperatures expected at redshifts $z < 500$.

However, as suggested by Peebles and Dicke (1968) and Saslaw and Zipoy (1967), trace amounts of H_2, formed from the H^- ion, could furnish sufficient cooling to trigger cloud collapse (Hutchins 1976), provided that the molecules remain undissociated by ultraviolet radiation or collisions with H atoms. Cooling is necessary for continued collapse, since the Jeans mass of a gravitationally unstable cloud will increase in adiabatic collapse.

In this paper, we examine molecule formation and spherical cloud collapse in the early universe. Our study differs from earlier work in three respects: (1) we include the molecules HD, LiH, and HeH^+, as well as H_2; (2) we adopt current, more accurate rates for gas-phase chemistry; and (3) we incorporate more accurate low-density rates for H_2 collisional dissociation during cloud collapse. We find that molecule formation is efficient following recombination, and that molecular cooling is effective in triggering the gravitational collapse of protostars or intergalactic shells.

II. MOLECULES DURING THE RECOMBINATION EPOCH

In the absence of dust grains, H_2 formation during the recombination epoch proceeds via two gas-phase mechanisms, commonly known as the H^- and the H_2^+ processes. Rates for the first process,

$$\begin{aligned} H + e^- &\rightarrow H^- + \gamma\,, \\ H^- + H &\rightarrow H_2 + e^-\,, \end{aligned} \tag{1}$$

have been computed by Dalgarno and Kingston (1963), de Jong (1972), and Bieniek and Dalgarno (1979), while those for the second process,

$$\begin{aligned} H + H^+ &\rightarrow H_2^+ + \gamma\,, \\ H_2^+ + H &\rightarrow H_2 + H^+\,, \end{aligned} \tag{2}$$

are given by Ramaker and Peek (1976) and Karpas, Anicich, and Huntress (1979). An additional channel for H_2 formation at high densities ($n > 10^9$ cm^{-3}) is the three-body process $3H \rightarrow H_2 + H$ (Palla, Salpeter, and Stahler 1983). Direct radiative association of $H + H \rightarrow H_2 + \gamma$ is highly forbidden, owing to the negligible dipole moment of the homonuclear H_2 molecule (Gould and Salpeter 1963).

However, HD and LiH possess finite dipole moments and may form by radiative association, through a radiative transition to a bound vibrational/rotational state in the ground electronic state. Lacking a quantal calculation of the rate coefficients, we estimate $R_{HX} = \sigma\bar{v}P_{rad}$, where $\sigma\bar{v} = 10^{-10}$ cm^3 s^{-1} is a typical collision rate coefficient and $P_{rad} = A_{tot}\,\Delta t$ is the probability that a radiative transition occurs during the collision interval Δt (Watson 1976). Here, A_{tot} is the total radiative decay rate from a quasi-bound state in the continuum to all bound states. We approximate $\Delta t = D_{eff}/\bar{v}$, $D_{eff} = (\sigma/\pi\bar{v})^{1/2}$, and $A_{tot} = A_1(E/\Delta E)^3 N_{tr}$, where A_1 is a typical radiative decay rate for a $\Delta v = 1$ vibrational transition of energy ΔE, E is a typical energy of the transition from the continuum to bound vibrational states, and N_{tr} is the total number of such transitions (approximately twice the number of bound vibrational states to which capture proceeds). Assuming vibrational A values for LiH (Docken and Hinze 1972) and HD (Abgrall, Roueff, and Viala 1983), $E = 50\%$ of the dissociation energy and $N_{tr} \approx 6$, we estimate that $R(HD) = 10^{-25}$ cm^3 s^{-1} and $R(LiH) = 10^{-17}$ cm^3 s^{-1}. Rates for HeH^+ are given by Roberge and Dalgarno (1982). We also include HD formation through the H^- and H_2^+ channels, at rates diminished by the cosmological D/H ratio.

The results of a detailed integration are shown in Figure 1. Table 1 lists the molecular dissociation energies E_d, radiative association rate coefficients R_{HX}, and integrated molecular abundances for three cosmological models. We assume

[1] Department of Physics, University of Colorado.
[2] Department of Astrophysics, University of Colorado; and Laboratory for Atmospheric and Space Physics.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1984ApJ...280..465L

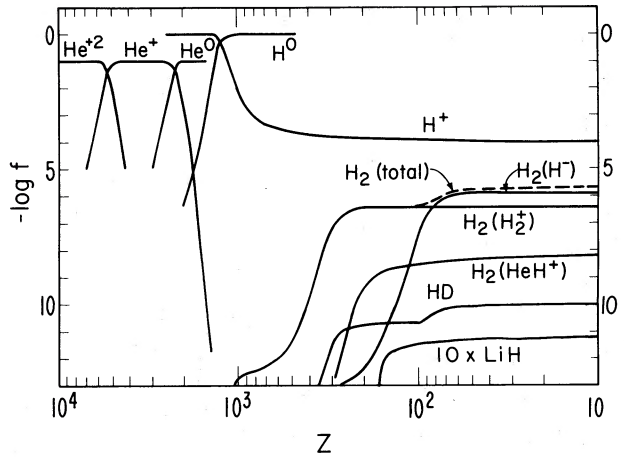


FIG. 1.—Abundance fractions of atoms, ions, and molecules vs. redshift in standard Friedmann cosmological model 1 ($\Omega_0 = \Omega_b = 0.1$, $h = 0.5$; see Table 1). Various channels for H_2 formation are indicated by the intermediate species in parentheses. The HD curve includes both radiative association and the isotropic versions of H_2 formation via H^- and H_2^+.

standard equations (Peebles 1971) for a Friedmann cosmology, with Hubble constant $H_0 = (100 \text{ km s}^{-1} \text{ Mpc}^{-1})h$, closure parameter Ω_0 (with and without hypothetical massive leptons), and present radiation temperature $T_0 = 3$ K. We follow the expansion rate, thermal history of matter and radiation, and ionization and molecular fractions during the recombination epoch, assuming H and He photoionization from the blackbody radiation, case B recombination, and Compton heating of the matter. Once Compton heating becomes less important than expansion cooling, the radiation and matter cool adiabatically as $T_r \approx (1 + z)$ and $T_m \approx (1 + z)^2$.

We integrate the molecular network involving H, H^-, H_2^+, H^+, and e^-, using both forward and reverse reactions in equations (1) and (2), and similar equations for HD, LiH, and HeH^+. Photodestruction of the various species is dominated by the cosmic background radiation; we have verified that the absorption rate of photons produced in reactions (1) and (2) is small. Lyα recombination photons may have a small effect on H^- at $z = 100$, but the modeling of the transfer is beyond the scope of this paper. The initial abundances of D, Li, and He are taken from recent calculations of Yang *et al.* (1984) for appropriate values of H_0 and Ω_0. For $\Omega_0 = 0.1$ and $h = 0.5$, $D/H = 5 \times 10^{-5}$, $Li/H = 10^{-10}$, $He/H = 0.1$, $H_2/H \approx 2 \times 10^{-6}$, $HD/H \approx 10^{-10}$, and $LiH/H \approx 10^{-11}$. The trace catalysts for H_2 formation reach peak abundances of $H^-/H \approx 10^{-11}$ ($z \approx 80$), $H_2^+/H \approx 10^{-12}$ ($z \approx 280$), and $HeH^+/H \approx 4 \times 10^{-14}$ ($z \approx 10$), then diminish as they are converted to H_2.

It is useful to define the redshift z_{eff} as the epoch at which the fractional abundance of molecule HX first becomes significant according to the Saha equation ($f_H f_X / f_{HX} \approx 1$). The H_2 forms primarily by the H^- process at $z_{\text{eff}} = 60$ and by the H_2^+ process at $z_{\text{eff}} = 200$. A small amount of H_2 is produced at $z_{\text{eff}} = 140$ from HeH^+, which charge exchanges with H to form H_2^+. The HeH^+ abundance is small, because the usual formation mode, radiative association of He^+ and H^0, is negligible owing to the small He^+ fraction. Thus, the slower mode, radiative association of He^0 and H^+ (Roberge and Dalgarno 1982) prevails.

The trace species HD and LiH form by radiative association near $z_{\text{eff}} = 300$ and 200, respectively. However, most of the HD forms through the isotopic versions of the H^- and H_2^+ processes. One can estimate the final abundance of each molecule by assuming that once its destruction mechanisms are inoperative (at z_{eff}), the molecule undergoes a burst of formation for a time $t_H(z_{\text{eff}})$, equal to the Hubble expansion time at that epoch. For $t \gg t_H(z_{\text{eff}})$, the density is much lower, and further formation is negligible, unless a cloud collapses gravitationally. This analytic approximation leads to the estimate,

$$(n_{HX}/n_H) = (n_X/n_H) R_{HX} n_H(z_{\text{eff}}) t_H(z_{\text{eff}}) , \quad (3)$$

where n_X/n_H is the cosmological abundance of atomic species X, R_{HX} the radiative association rate coefficient, and $t_H = H_0^{-1}(1 + z)^{-1}(1 + \Omega_0 z)^{-1/2}$ the Hubble expansion time. The hydrogen density is given by $n_H = (8.02 \times 10^{-6} \text{ cm}^{-3}) \times \Omega_b h^2 (1 + z)^3$, where Ω_b is the closure parameter in baryons and where $He/H = 0.1$. Our numerical results for HD and LiH are in good agreement with this estimate. Equation (3) may also be applied to H_2 formation by calculating the

TABLE 1

MOLECULAR DATA AND ABUNDANCES

MOLECULE[a]	E_d[b] (eV)	R_{HX}[c] ($\text{cm}^3 \text{ s}^{-1}$)	z_{eff}[d]	FRACTIONAL ABUNDANCE[e] Model 1	Model 2	Model 3
H_2 (via H^-)	(0.75)	1.0(−16)	64	1.4(−6)	8.3(−7)	1.3(−6)
H_2 (via H_2^+)	(2.65)	1.4(−18)	190	3.8(−7)	3.5(−7)	3.6(−7)
H_2 (via HeH^+)	(1.96)	4.1(−18)	140	5.4(−9)	5.1(−9)	6.1(−9)
H_2 (total)	...	...	...	1.7(−6)	1.2(−6)	1.7(−6)
HD (radiative)	4.48	1.0(−25)	330	1.1(−14)	3.5(−15)	1.8(−13)
HD (total)	...	...	...	8.5(−11)	6.0(−11)	1.7(−11)
LiH	2.45	1.0(−17)	180	5.5(−13)	1.8(−13)	1.0(−11)

[a] Molecule HX or intermediate species used to form H_2.
[b] Dissociation energy of HX or (intermediate species).
[c] Rate coefficient for radiative association of HX or intermediate species, evaluated at $T(z_{\text{eff}})$.
[d] Redshift of formation ($f_H f_X / f_{HX} = 1$ in Saha equilibrium) in model 1.
[e] Computed abundance fractions, n_{HX}/n_H, for various formation mechanisms. Friedmann cosmological models assume: model 1 ($\Omega_0 = \Omega_b = 0.1$, $h = 0.5$); model 2 ($\Omega_b = 0.1$, $\Omega_0 = 1.0$, $h = 0.5$—includes leptonic matter); model 3 ($\Omega_0 = \Omega_b = 0.3$, $h = 0.5$). Initial atomic abundances: models 1 and 2 ($D/H = 5 \times 10^{-5}$, $Li/H = 10^{-10}$); model 3 ($D/H = 10^{-5}$, $Li/H = 10^{-9}$).

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1984ApJ...280..465L

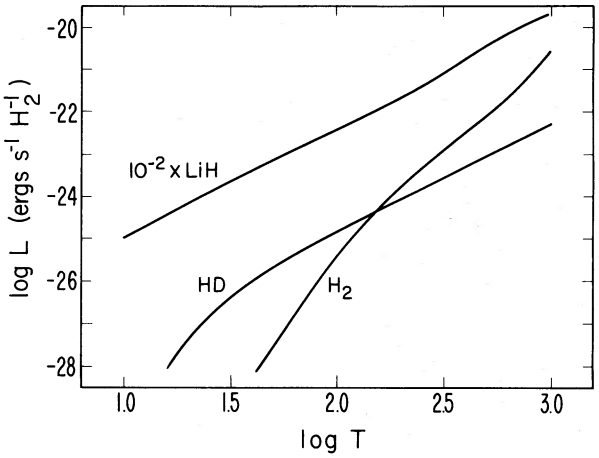


FIG. 2.—Cooling rate per H_2 molecule for H_2, HD, and LiH rotational and vibrational states in thermal equilibrium (abundances $HD/H_2 = 10^{-4}$, $LiH/H_2 = 10^{-6}$). Both HD and LiH are important at low temperatures, although at low densities ($n \ll 10^7$ and 10^{12} cm^{-3} for HD and LiH, respectively) the underpopulation of rotational levels relative to Boltzmann values will diminish $L(T)$ from the rates plotted. Note that LiH cooling has been multiplied by 10^{-2}.

H^- or H_2^+ formed at the appropriate $z_{\rm eff}$ (see Table 1), and by assuming that they are entirely converted to H_2.

III. MOLECULAR COOLING AND COLD COLLAPSE

Because of its rotational and vibrational levels, H_2 is an important coolant of interstellar gas from 10^2 to 10^4 K (Shull and Beckwith 1982). The trace molecules HD and LiH are also important, since their allowed dipole rotational transitions provide cooling to temperatures well below that of the lowest H_2 quadrupole transition, $E(J = 2) = 510$ K (Dalgarno and Wright 1972). Furthermore, HD and LiH dissociate far more slowly at low density than H_2, owing to radiative stabilization of their excited vibrational states (Dalgarno and Roberge 1979; Lepp and Shull 1983). The onset of high dissociation rates from a Boltzmann population of vibrational levels occurs when the H density (cm^{-3}) exceeds 10^5(H$_2$), 10^7(HD), and 10^{12}(LiH). Consequently, HD and LiH will survive longer than H_2 as coolants in protostellar collapse.

In order for a cloud to cool more rapidly than its free-fall time, its molecular fraction, $f = n(H_2)/n_H$, must be greater than

$$f(H_2) > (\tfrac{3}{2})(32G\rho/3\pi)^{1/2}kT/L(H_2)\,, \qquad (4)$$

where ρ is the initial mass density, T the gas temperature, and $L(H_2)$ the total cooling rate (ergs s^{-1} per H_2). For a Jeans unstable cloud at $z = 50$, collapsing adiabatically to 500 K and $n_H \approx 1$ cm^{-3}, $f(H_2) > 10^{-6}$ is sufficient to radiate away the gravitational energy. Figure 2 illustrates the cooling rates of H_2, LiH, and HD under conditions of thermal equilibrium. Because Boltzmann level populations are not immediately reached during the collapse of low-density clouds, the initial HD and LiH cooling will be smaller, and a more detailed calculation is necessary to follow the excitation and radiative decay of the molecular rotational and vibrational states.

To illustrate these details, we have computed a model of a homogeneous cloud undergoing spherical, self-similar, free-fall collapse. In order to neglect pressure gradients, we assume that the cloud mass M is several times the local Jeans mass M_J. The equations of motion yield the compression factor $F = (n/n_0)$ for all spherical shells as a function of time. For an initially homogeneous medium, each collapsing shell has the same free-fall time; thus, a single temperature and density characterize the entire cloud until pressure gradients or opacity become important. We start the integration with the molecular fractions described in Table 1 and with $T_0 = 64$ K and $n_0 = 0.027$ cm^{-3}, corresponding to conditions at $z = 50$ ($\Omega_0 = 0.1$, $h = 0.5$). The Jeans length $\lambda_J \approx 560$ pc and the Jeans mass $M_J \approx 10^5\ M_\odot$, making this object similar to protoglobular clusters. Rates for H_2 cooling and collisional dissociation were taken from Lepp and Shull (1983). The cooling and dissociation rates for HD and LiH were computed directly, assuming the same excitation cross sections as H_2 but increasing the radiative decay rates appropriately.

Figure 3 shows the temperature-density track of the collapse, together with the molecular and ionized fractions. Figure 4 illustrates the relative importance of cooling by H_2, HD, and LiH. At $F = 10^{12}$, both H_2 and HD are cooling at their high density limits (Boltzman level populations). However, the hydrogen density is still a factor of 50 below that needed to thermalize LiH level populations. Thus, LiH will carry an increasingly large fraction of the cooling for $F > 10^{12}$. In addition, the LiH rotational lines remain optically thin much longer than those of H_2 or HD, as discussed below. We conclude that LiH cooling may dominate the later stages of collapse.

Because of the efficient molecular cooling, the temperature remains below 10^3 K during most of the collapse. The H_2/H abundance rises to 10^{-3} at a compression $F = 10^7$, leveling off as H^- catalysis ceases. Nearly all the Li goes into LiH by $F = 10^{11}$. The moderate temperatures ensure that collisional dissociation is unimportant at this stage. When H_2 finally dissociates ($T \geq 4000$ K), it does so at its high density limit (thermal level populations).

When $n_H > 10^9$ cm^{-3} ($F \approx 10^{11}$), three-body H_2 formation and opacity effects must be considered. Three-body formation

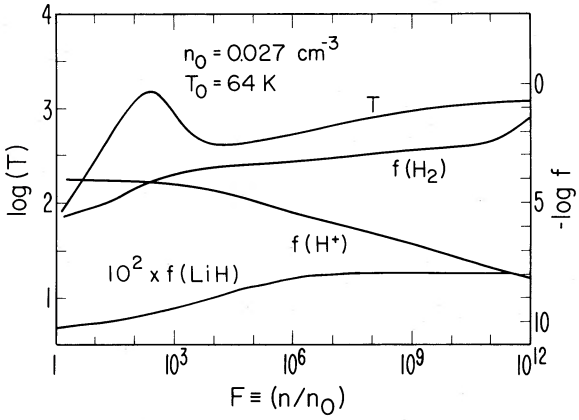


FIG. 3.—Temperature and density track of spherical cloud in self-similar free-fall collapse at redshift $z = 50$. Fractional H^+ and H_2 abundances are plotted at right. Molecular cooling causes temperature to depart from adiabatic value ($T \propto n^{2/3}$) at 10^3 K, and the formation of additional molecules in the collapse cools cloud below 500 K until $n > 10^9$ cm^{-3}, when three-body H_2 formation sets in.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1984ApJ...280..465L

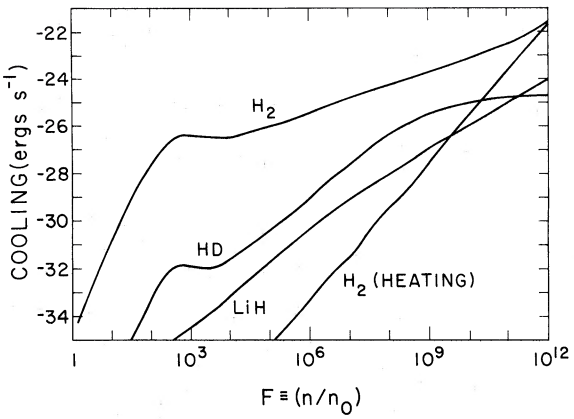


FIG. 4.—Relative cooling rates by H_2, HD, and LiH (ergs s^{-1}) during cloud collapse (same model as Fig. 3). Heating by three-body H_2 formation becomes important as F approaches 10^{12} ($n \approx 3 \times 10^{10}$ cm^{-3}). Although LiH cooling is still relatively small at $F = 10^{12}$, it will become increasingly more important at high densities, as described in § III.

causes a sudden rise in the abundance of H_2, and the heat released by this formation produces a slight rise in the temperature of the cloud (see Figs. 3 and 4). In our model, H^- never becomes an important opacity source, because its abundance drops as the electrons recombine at high density. The cooling rotational lines become optically thick at column densities:

$$N(\tau = 1) = \begin{cases} (10^{24}\ \text{cm}^{-2})\Delta v & \text{for } H_2 \\ (10^{20}\ \text{cm}^{-2})\Delta v & \text{for HD} \\ (10^{15}\ \text{cm}^{-2})\Delta v & \text{for LiH} \end{cases}, \quad (5)$$

where Δv (km s^{-1}) is the velocity width of the lines induced by cloud collapse. Equating Δv to the bulk motion of the infalling shells, we estimate that $\Delta v \approx 100$ km s^{-1} at $F = 10^{12}$. Although this velocity is clearly supersonic, it is not obvious that the collapsing flow will become turbulent or subject to shocks. Such effects might not arise until the cooling lines become optically thick, so that pressure waves can propagate upstream (outward through the cloud). In our collapse model, the H_2 column density through the cloud is $N(H_2) \approx (5 \times 10^{19}\ \text{cm}^{-2}) f(H_2) F^{2/3}$. Since $HD/H_2 \approx 10^{-4}$, the HD and H_2 lines become optically thick at the same point, as three-body formation sets in ($F \approx 10^{12}$, $f \approx 0.1$). However, since $Li/H \approx 10^{-10}$, the LiH lines remain optically thin and effective coolants until much higher densities. Our results are in agreement with those of Palla, Salpeter, and Stahler (1983), who pointed out the importance of three-body H_2 formation in primeval star formation. When the cloud has collapsed this far, the effects of rotation or possible magnetic fields will almost certainly dominate. We will discuss rotational support, spheroidal collapse, and molecular opacity in a later paper.

IV. DISCUSSION

These calculations show that molecule formation at high redshift is effective, and that the resulting cooling can have important effects on the gravitational collapse and thermal state of intergalactic gas clouds at $z < 200$. At $z < 60$, several parts per million of H are in the form of H_2, while $HD/H_2 \approx 10^{-4.5}$ and $LiH/H_2 \approx 10^{-6.5}$. Our H_2 abundances are similar to those computed by previous workers (Takeda, Sato, and Matsuda 1969; Hirasawa, Aizu, and Taketani 1969), while our results on HD, LiH, and HeH^+ are new.

The amounts of HD and LiH formed by radiative association depend almost linearly on the adopted rate coefficients (10^{-25} and 10^{-17} cm^3 s^{-1}, respectively). Detailed quantal calculations would be valued to check the validity of our estimates. It is also possible that D/H and Li/H differ from the values predicted by nucleosynthesis in an open universe ($\Omega_b \approx 0.1$). For example, Vidal-Madjar *et al.* (1983) suggest that interstellar D/H could be as low as 5×10^{-6}, characteristic of $\Omega_b \geq 0.3$. In such a cosmology, Li/H would be 10^{-9}, and LiH cooling would be extremely important (model 3, Table 1). However, Spite and Spite (1982) find $Li/H \approx 10^{-10}$ in old halo stars.

The computed molecular abundances are sufficient to trigger and maintain gravitational collapse of Jeans unstable clouds through radiative cooling. As the H_2 abundance rises in the collapse, radiative cooling accelerates and T drops below 500 K; H_2 collisional dissociation is negligible. Since the isothermal Jeans mass decreases as $n^{-1/2}$ with increasing density, fragmentation and thermal instability are likely (Silk 1983). It is entirely possible that low-mass stars could form during the first generation of cloud collapse. In collapse at $z = 50$, $\sim 3 \times 10^{47}$ ergs $M_\odot^{-1}$ of gravitational energy is radiated away. For 10^{11} $M_\odot$ of protostars, this energy, spread over a free-fall collapse time, corresponds to $\sim 6 \times 10^8$ $L_\odot$ in H_2 lines between 2 and 10 μm.

Molecular coolants are also important for intergalactic blast waves (Ostriker and Cowie 1981; Shull and Silk 1981). In the absence of dissociating ultraviolet radiation, rotational excitation of H_2 can provide cooling down to temperatures ~ 100 K. The trace species HD and LiH could, in principle, cool the gas to even lower temperatures, although the time scale at low densities ($n < 1$ cm^{-3}) may be prohibitively long. If intergalactic gas shells are unstable to gravitational collapse, low postshock temperatures are conducive to the formation of cluster-sized objects (Ostriker 1983). However, for the molecular coolants to survive dissociation in hot postshock gas, the shells must have slowed below 35–50 km s^{-1} (Hollenbach and McKee 1980), limiting this scenario to fairly recent epochs.

We thank A. Dalgarno, R. McCray, J. Ostriker, J. Peebles, and E. Zweibel for valuable discussions during the course of this work and at the Aspen Center for Physics. Our research was supported by NSF grant AST82-16481 at the University of Colorado.

REFERENCES

Abgrall, H., Roueff, E., and Viala, Y. 1982, *Astr. Ap. Suppl.*, **50**, 505.
Bieniek, R. J., and Dalgarno, A. 1979, *Ap. J.*, **228**, 635.
Dalgarno, A., and Kingston, A. E. 1963, *Observatory*, **83**, 39.
Dalgarno, A., and Roberge, W. 1979, *Ap. J. (Letters)*, **233**, L5.
Dalgarno, A., and Wright, E. L. 1972, *Ap. J. (Letters)*, **174**, L49.
de Jong, T. 1972, *Astr. Ap.*, **20**, 263.
Docken, K., and Hinze, J. 1972, *J. Chem. Phys.*, **57**, 4936.
Gould, R. J., and Salpeter, E. E. 1963, *Ap. J.*, **138**, 393.
Hirasawa, T., Aizu, K., and Taketani, M. 1969, *Progr. Theor. Phys.*, **41**, 835.
Hollenbach, D. J., and McKee, C. F. 1980, *Ap. J. (Letters)*, **241**, L47.
Hutchins, J. B. 1976, *Ap. J.*, **205**, 103.
Karpas, Z., Anicich, V., and Huntress, W. 1979, *J. Chem. Phys.*, **70**, 2877.
Lepp, S., and Shull, J. M. 1983, *Ap. J.*, **270**, 578.
Ostriker, J. P. 1983, private communication.

© American Astronomical Society • Provided by the NASA Astrophysics Data System

1984ApJ...280..465L

Ostriker, J. P., and Cowie, L. L. 1981, *Ap. J. (Letters)*, **243**, L127.
Palla, F., Salpeter, E. E., and Stahler, S. W. 1983, *Ap. J.*, **271**, 632.
Peebles, P. J. E. 1966, *Ap. J.*, **146**, 542.
———. 1971, *Physical Cosmology* (Princeton: Princeton University Press).
Peebles, P. J. E., and Dicke, R. H. 1968, *Ap. J.*, **154**, 891.
Ramaker, D. E., and Peek, J. M. 1966, *Phys. Rev. A*, **13**, 58.
Roberge, W., and Dalgarno, A. 1982, *Ap. J.*, **255**, 489.
Saslaw, W. C., and Zipoy, D. 1967, *Nature*, **216**, 976.
Shull, J. M., and Beckwith, S. 1982, *Ann. Rev. Astr. Ap.*, **20**, 163.
Shull, J. M., and Silk, J. 1981, *Ap. J.*, **249**, 26.
Silk, J. 1983, *M.N.R.A.S.*, **205**, 705.
Spite, M., and Spite, F. 1982, *Astr. Ap.*, **115**, 357.
Takeda, H., Sato, H., and Matsuda, T. 1969, *Progr. Theor. Phys.*, **41**, 840.
Vidal-Madjar, A., Laurent, C., Gry, C., Bruston, P., Ferlet, R., and York, D. G. 1983, *Astr. Ap.*, **120**, 58.
Wagoner, R. V., Fowler, W. A., and Hoyle, F. 1967, *Ap. J.*, **148**, 3.
Watson, W. D. 1976, *Rev. Mod. Phys.*, **48**, 513.
Yang, J., Turner, M. S., Steigman, G., Schramm, D. N., and Olive, K. A. 1984, *Ap. J.*, **281**, in press.

STEPHEN LEPP: Center for Astrophysics, 60 Garden St., Cambridge, MA 02138

J. MICHAEL SHULL: Joint Institute for Laboratory Astrophysics, Campus Box 440, University of Colorado, Boulder, CO 80309

© American Astronomical Society • Provided by the NASA Astrophysics Data System